Data Base System

Muhammad Adil Saeed | 24k-0705 | Lab 7

IN-LAB TASK:

1.

```
SELECT E.FIRST_NAME, E.LAST_NAME, J.JOB_TITLE, D.DEPARTMENT_NAME, E.SALARY
FROM EMPLOYEES E
JOIN JOBS J
ON E.JOB_ID = J.JOB_ID
JOIN DEPARTMENTS D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
ORDER BY D.DEPARTMENT_NAME ASC;
```

Query Result

SQL | Fetched 50 rows in 0.003 seconds

	FIRST_NAME	LAST_NAME	JOB_TITLE	DEPARTMENT_NAME	SALARY
1	William	Gietz	Public Accountant	Accounting	8300
2	Shelley	Higgins	Accounting Manager	Accounting	15982.65
3	Jennifer	Whalen	Administration Assistant	Administration	5856.4
4	Neena	Kochhar	Administration Vice President	Executive	5000
5	Adnan khan	King	President	Executive	5000
6	Lex	De Haan	Administration Vice President	Executive	5000
7	Nancy	Greenberg	Finance Manager	Finance	15982.65
8	Jose Manuel	Urman	Accountant	Finance	7800

2.

```
SELECT D.DEPARTMENT_ID, D.DEPARTMENT_NAME, COUNT(E.EMPLOYEE_ID) AS TOTAL_EMPLOYEES
FROM DEPARTMENTS D
JOIN EMPLOYEES E
ON D.DEPARTMENT_ID = E.DEPARTMENT_ID
GROUP BY D.DEPARTMENT_ID, D.DEPARTMENT_NAME
HAVING COUNT(E.EMPLOYEE_ID) >= 2;
```

Query Result

SQL | All Rows Fetched: 8 in 0.002 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	TOTAL_EMPLOYEES
1	100	Finance	6
2	50	Shipping	45
3	30	Purchasing	6
4	90	Executive	3
5	110	Accounting	2
6	20	Marketing	2
7	60	IT	5
8	80	Sales	34

3.

```
SELECT E.FIRST_NAME, E.LAST_NAME, J.JOB_TITLE,
       (E.SALARY*12) AS ANNUAL_SALARY,
       D.DEPARTMENT_NAME, L.CITY
FROM EMPLOYEES E
JOIN JOBS J
ON E.JOB_ID = J.JOB_ID
JOIN DEPARTMENTS D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
JOIN LOCATIONS L
ON D.LOCATION_ID = L.LOCATION_ID
WHERE (E.SALARY*12) >= 60000
AND J.JOB_TITLE <> 'ANALYST';
```

Query Result

SQL | Fetched 50 rows in 0.003 seconds

	FIRST_NAME	LAST_NAME	JOB_TITLE	ANNUAL_SALARY	DEPARTMENT_NAME	CITY
1	Alexander	Hunold	Programmer	108000	IT	Southlake
2	Bruce	Ernst	Programmer	72000	IT	Southlake
3	Payam	Kaufling	Stock Manager	94800	Shipping	South San Francisco
4	Matthew	Weiss	Stock Manager	96000	Shipping	South San Francisco
5	Adam	Fripp	Stock Manager	98400	Shipping	South San Francisco
6	Shanta	Vollman	Stock Manager	78000	Shipping	South San Francisco
7	Kevin	Mourgos	Stock Manager	69600	Shipping	South San Francisco
8	William	Gietz	Public Accountant	99600	Accounting	Seattle
9	Shelley	Higgins	Accounting Manager	191791.8	Accounting	Seattle
10	Daniel	Faviet	Accountant	108000	Finance	Seattle
11	John	Chen	Accountant	98400	Finance	Seattle

4.

```
SELECT DISTINCT M.*
FROM EMPLOYEES E
JOIN EMPLOYEES M
ON E.MANAGER_ID = M.EMPLOYEE_ID;
```

Query Result

SQL | All Rows Fetched: 18 in 0.004 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
2	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	5000	0.4	100	80
3	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	5000	0.2	100	80
4	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	5000	(null)	100	90
5	205	Shelley	Higgins	SHIGGINS	515.123.8080	07-JUN-02	AC_MGR	15982.65	(null)	101	110
6	100	Adnan khan	King	SKING	515.123.4567	17-JUN-03	AD_PRES	5000	(null)	(null)	90
7	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
8	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
9	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	15982.65	(null)	101	100
10	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
11	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
12	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	5000	0.3	100	80
13	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	5000	0.3	100	80
14	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
15	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	5000	0.3	100	80
16	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20
17	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	5000	(null)	100	90
18	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60

5.

```
SELECT D.DEPARTMENT_ID, D.DEPARTMENT_NAME
FROM DEPARTMENTS D
LEFT JOIN EMPLOYEES E
ON D.DEPARTMENT_ID = E.DEPARTMENT_ID
WHERE E.EMPLOYEE_ID IS NULL;
```

Query Result

SQL | All Rows Fetched: 16 in 0.003 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME
1	120	Treasury
2	130	Corporate Tax
3	140	Control And Credit
4	150	Shareholder Services
5	160	Benefits
6	170	Manufacturing
7	180	Construction

6.

```
SELECT E.FIRST_NAME, E.LAST_NAME, E.SALARY, D.DEPARTMENT_NAME
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

Query Result

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_NAME
1	Jennifer	Whalen	5856.4	Administration
2	Pat	Fay	6000	Marketing
3	Michael	Hartstein	13000	Marketing
4	Karen	Colmenares	2500	Purchasing
5	Guy	Himuro	2600	Purchasing
6	Sigal	Tobias	2800	Purchasing
7	Shelli	Baida	2900	Purchasing
8	Alexander	Khoo	3100	Purchasing
9	Den	Raphaely	11000	Purchasing
10	Susan	Mavris	8651.5	Human Resources
11	Douglas	Grant	2600	Shipping
12	Donald	OConnell	2600	Shipping

7.

```
SELECT E.FIRST_NAME, E.LAST_NAME, J.JOB_TITLE,
       D.DEPARTMENT_NAME, L.CITY
FROM EMPLOYEES E
JOIN JOBS J
ON E.JOB_ID = J.JOB_ID
JOIN DEPARTMENTS D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
JOIN LOCATIONS L
ON D.LOCATION_ID = L.LOCATION_ID
WHERE L.STATE_PROVINCE IS NULL;
```

Query Result

SQL | All Rows Fetched: 1 in 0.003 seconds

	FIRST_NAME	LAST_NAME	JOB_TITLE	DEPARTMENT_NAME	CITY
1	Susan	Mavris	Human Resources Representative	Human Resources	London

8.

```
SELECT EMPLOYEE_ID
FROM EMPLOYEES
MINUS
SELECT EMPLOYEE_ID
FROM JOB_HISTORY;
```

Query Result

SQL | Fetched 50 rows in 0.005 seconds

	EMPLOYEE_ID
1	100
2	103
3	104
4	105
5	106
6	107

9.

```
CREATE VIEW EMP_DEPT_VIEW AS
SELECT E.FIRST_NAME, E.LAST_NAME, D.DEPARTMENT_NAME
FROM EMPLOYEES E
JOIN DEPARTMENTS D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

Query Result × | Script Output ×

Task completed in 0.032 seconds

```
View EMP_DEPT_VIEW created.
```

10.

```
CREATE OR REPLACE VIEW EMP_DEPT_VIEW AS
SELECT E.FIRST_NAME, E.LAST_NAME, D.DEPARTMENT_NAME, E.SALARY
FROM EMPLOYEES E
JOIN DEPARTMENTS D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

Query Result × | Script Output ×

Task completed in 0.033 seconds

```
View EMP_DEPT_VIEW created.

View EMP_DEPT_VIEW created.
```

POST-LAB TASK:

1.

```
SELECT DISTINCT E.EMPLOYEE_ID, E.FIRST_NAME, E.LAST_NAME
FROM EMPLOYEES E
JOIN JOB_HISTORY JH
ON E.EMPLOYEE_ID = JH.EMPLOYEE_ID;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 7 in 0.002 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME
1	101	Neena	Kochhar
2	200	Jennifer	Whalen
3	102	Lex	De Haan
4	201	Michael	Hartstein
5	114	Den	Raphaely
6	122	Payam	Kaufling
7	176	Jonathon	Taylor

2.

```
SELECT E.FIRST_NAME AS EMPLOYEE,
       M.FIRST_NAME AS MANAGER,
       E.DEPARTMENT_ID
FROM EMPLOYEES E
JOIN EMPLOYEES M
ON E.MANAGER_ID = M.EMPLOYEE_ID
WHERE E.DEPARTMENT_ID = M.DEPARTMENT_ID;
```

Script Output × | Query Result ×

SQL | Fetched 50 rows in 0.002 seconds

	EMPLOYEE	MANAGER	DEPARTMENT_ID
1	Lex	Adnan khan	90
2	Neena	Adnan khan	90
3	Diana	Alexander	60
4	Valli	Alexander	60
5	David	Alexander	60
6	Bruce	Alexander	60
7	Luis	Nancy	100
8	Jose Manuel	Nancy	100
9	Ismael	Nancy	100
10	John	Nancy	100
11	Daniel	Nancy	100
12	Karen	Den	30

3.

```
SELECT FIRST_NAME, LAST_NAME, DEPARTMENT_ID
FROM EMPLOYEES
WHERE DEPARTMENT_ID IN
      (SELECT DEPARTMENT_ID
       FROM EMPLOYEES
       GROUP BY DEPARTMENT_ID
       HAVING COUNT(*) > 5);
```

Script Output × | Query Result ×

SQL | Fetched 50 rows in 0.003 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
1	Jose Manuel	Urman	100
2	Guy	Himuro	30
3	Adam	Fripp	50
4	TJ	Olson	50
5	Jason	Mallin	50
6	Hazel	Philtanker	50
7	Peter	Vargas	50
8	Karen	Partners	80
9	David	Bernstein	80
10	Peter	Hall	80

4.

```
SELECT E.FIRST_NAME AS EMPLOYEE,
       M.FIRST_NAME AS MANAGER
FROM EMPLOYEES E
LEFT JOIN EMPLOYEES M
ON E.MANAGER_ID = M.EMPLOYEE_ID;
```

Script Output × | Query Result ×

SQL | Fetched 50 rows in 0.003 seconds

	EMPLOYEE	MANAGER
1	Sundita	Gerald
2	Elizabeth	Gerald
3	William	Gerald
4	Tayler	Gerald
5	Harrison	Gerald
6	Lisa	Gerald
7	Alexander	Lex
8	Amit	Alberto
9	Sundar	Alberto
10	David	Alberto

5.

```
SELECT FIRST_NAME, LAST_NAME, SALARY
FROM EMPLOYEES E
WHERE SALARY >
      (SELECT AVG(SALARY)
       FROM EMPLOYEES
       WHERE DEPARTMENT_ID = E.DEPARTMENT_ID)
AND SALARY <
      (SELECT MAX(SALARY)
       FROM EMPLOYEES);
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 17 in 0.007 seconds

	FIRST_NAME	LAST_NAME	SALARY
1	Den	Raphaely	11000
2	Michael	Hartstein	13000
3	Britney	Everett	3900
4	Sarah	Bell	4000
5	Jennifer	Dilly	3600
6	Kelly	Chung	3800
7	Alexis	Bull	4100

6.

```
SELECT E.FIRST_NAME, E.LAST_NAME,
       D.DEPARTMENT_NAME,
       JH.START_DATE, JH.END_DATE
FROM EMPLOYEES E
JOIN JOB_HISTORY JH
ON E.EMPLOYEE_ID = JH.EMPLOYEE_ID
JOIN DEPARTMENTS D
ON JH.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 10 in 0.004 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME	START_DATE	END_DATE
1	Neena	Kochhar	Accounting	21-SEP-97	27-OCT-01
2	Neena	Kochhar	Accounting	28-OCT-01	15-MAR-05
3	Lex	De Haan	IT	13-JAN-01	24-JUL-06
4	Den	Raphaely	Shipping	24-MAR-06	31-DEC-07
5	Payam	Kaufling	Shipping	01-JAN-07	31-DEC-07
6	Jonathon	Taylor	Sales	24-MAR-06	31-DEC-06
7	Jonathon	Taylor	Sales	01-JAN-07	31-DEC-07
8	Jennifer	Whalen	Executive	17-SEP-95	17-JUN-01
9	Jennifer	Whalen	Executive	01-JUL-02	31-DEC-06
10	Michael	Hartstein	Marketing	17-FEB-04	19-DEC-07

7.

```
SELECT E.EMPLOYEE_ID, E.FIRST_NAME, E.LAST_NAME
FROM EMPLOYEES E
JOIN JOB_HISTORY JH
ON E.EMPLOYEE_ID = JH.EMPLOYEE_ID
JOIN DEPARTMENTS D
ON JH.DEPARTMENT_ID = D.DEPARTMENT_ID
JOIN LOCATIONS L
ON D.LOCATION_ID = L.LOCATION_ID
GROUP BY E.EMPLOYEE_ID, E.FIRST_NAME, E.LAST_NAME
HAVING COUNT(DISTINCT L.COUNTRY_ID) > 1;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 0 in 0.008 seconds

EMPLOYE...	FIRST_NA...	LAST_NAME

8.

```
SELECT E1.FIRST_NAME, E1.LAST_NAME, E1.DEPARTMENT_ID,
       E2.FIRST_NAME, E2.DEPARTMENT_ID
FROM EMPLOYEES E1
JOIN EMPLOYEES E2
ON E1.LAST_NAME = E2.LAST_NAME
AND E1.DEPARTMENT_ID <> E2.DEPARTMENT_ID
AND E1.EMPLOYEE_ID <> E2.EMPLOYEE_ID;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 4 in 0.001 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_ID	FIRST_NAME_1	DEPARTMENT_ID_1
1	Janette	King	80	Adnan khan	90
2	Adnan khan	King	90	Janette	80
3	Winston	Taylor	50	Jonathon	80
4	Jonathon	Taylor	80	Winston	50

9.

```
SELECT FIRST_NAME, LAST_NAME, DEPARTMENT_ID, SALARY
FROM
(
SELECT FIRST_NAME, LAST_NAME, DEPARTMENT_ID, SALARY,
       DENSE_RANK() OVER
       (PARTITION BY DEPARTMENT_ID ORDER BY SALARY DESC) RANK_NO
FROM EMPLOYEES
)
WHERE RANK_NO <= 3;
```

Script Output × Query Result ×

SQL | Fetched 50 rows in 0.003 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_ID	SALARY
1	Jennifer	Whalen	10	5856.4
2	Michael	Hartstein	20	13000
3	Pat	Fay	20	6000
4	Den	Raphaely	30	11000
5	Alexander	Khoo	30	3100
6	Shelli	Baida	30	2900
7	Susan	Mavris	40	8651.5
8	Adam	Fripp	50	8200
9	Matthew	Weiss	50	8000
10	Payam	Kaufling	50	7900
11	Alexander	Hunold	60	9000
12	Bruce	Ernst	60	6000
13	David	Austin	60	4800

10.

```
CREATE VIEW HIGH_SALARY_EMP AS
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY
FROM EMPLOYEES
WHERE SALARY > 10000;
SELECT * FROM HIGH_SALARY_EMP;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 5 in 0.002 seconds

	EMPLOY...	FIRST_NAME	LAST_NAME	SALARY
1	108	Nancy	Greenberg	15982.65
2	114	Den	Raphaely	11000
3	201	Michael	Hartstein	13000
4	204	Hermann	Baer	13310
5	205	Shelley	Higgins	15982.65

11.

```
CREATE VIEW EMP_HIRED_AFTER_2005 AS
SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, HIRE_DATE
FROM EMPLOYEES
WHERE HIRE_DATE > TO_DATE('31-DEC-2005','DD-MON-YYYY');
SELECT * FROM EMP_HIRED_AFTER_2005;
```

Script Output | Query Result

Close Fetched 50 rows in 0.002 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	HIRE_DATE
1	103	Alexander	Hunold	03-JAN-06
2	104	Bruce	Ernst	21-MAY-07
3	106	Valli	Pataballa	05-FEB-06
4	107	Diana	Lorentz	07-FEB-07
5	112	Jose Manuel	Urman	07-MAR-06
6	113	Luis	Popp	07-DEC-07
7	118	Guy	Himuro	15-NOV-06
8	119	Karen	Colmenares	10-AUG-07
9	124	Kevin	Mourgos	16-NOV-07
10	126	Irene	Mikkilineni	28-SEP-06
11	127	James	Landry	14-JAN-07
12	128	Steven	Markle	08-MAR-08